

```
from google.colab import files
```

```
uploaded = files.upload()
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import StandardScaler
```

```
from sklearn.metrics import r2_score
```

```
from keras.models import Sequential
```

```
from keras.layers import Dense
```

```
# Load dataset
```

```
df = pd.read_csv('BostonHousing.csv')
```

```
# Display first 10 rows
```

```
print(df.head(10))
```

```
# Check null values
```

```
print(df.isnull().sum())
```

```
# Dataset information
```

```
print(df.info())
```

```
# Statistical summary
```

```
print(df.describe())
```

```
# Correlation with target column
```

```
print(df.corr()['medv'].sort_values())
```

```
# Features and target
```

```
X = df.loc[:, df.columns != 'medv'].values
```

```
y = df.loc[:, 'medv'].values
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X,
```

```
    y,
```

```
    test_size=0.25,
```

```
    random_state=42
```

```
)
```

```
# Standardization
```

```
scaler = StandardScaler()
```

```
scaler.fit(X_train)
```

```
X_train = scaler.transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
# Build ANN model
```

```
model = Sequential([
```

```
    Dense(128, input_shape=(13,)), activation='relu', name='dense_1'),
```

```
    Dense(64, activation='relu', name='dense_2'),
```

```
    Dense(1, activation='linear', name='dense_output')
```

```
])
```

```
# Compile model
```

```
model.compile(
```

```
    optimizer='adam',
```

```
    loss='mse',
```

```
    metrics=['mae']
)

# Model summary
model.summary()

# Train model
history = model.fit(
    X_train,
    y_train,
    epochs=100,
    validation_split=0.05,
    verbose='auto'
)

# Predictions
y_pred = model.predict(X_test)

# Evaluate model
mse_nn, mae_nn = model.evaluate(X_test, y_test)

# R2 score
r2 = r2_score(y_test, y_pred)

print('Mean squared error on test data:', mse_nn)
print('Mean absolute error on test data:', mae_nn)
print('R2 score:', r2)

# Plot actual vs predicted values
plt.figure(figsize=(10, 10))
```

```
plt.scatter(y_test, y_pred, c='green')
```

```
p1 = max(max(y_pred), max(y_test))
```

```
p2 = min(min(y_pred), min(y_test))
```

```
plt.plot([p1, p2], [p1, p2], 'b-')
```

```
plt.xlabel('True Values', fontsize=15)
```

```
plt.ylabel('Predictions', fontsize=15)
```

```
plt.axis('equal')
```

```
plt.show()
```